

$$\textcircled{8} \quad L = \{ a^i b^j c^k d^m \mid i+k = j+m \text{ o } i+j+k=m \quad i,j,k,m \geq 0 \}$$

$$S \rightarrow E/F$$

$$\left. \begin{array}{l} E \rightarrow aEd \mid \lambda \\ J \rightarrow bJd \mid K \\ K \rightarrow cKd \mid \lambda \end{array} \right\} i+j+k=m$$

$$\left. \begin{array}{l} F \rightarrow aFd \mid LPM \\ L \rightarrow aLb \mid \lambda \\ M \rightarrow cMd \mid \lambda \\ P \rightarrow bPc \mid \lambda \end{array} \right\} i+k=j+m$$

Paso 1: Eliminar símbolos inútiles

$$\forall S \in G$$

S es generador

S es alcanzable

Paso 2: Eliminar proyecciones con λ

$$F \rightarrow ad \mid aFd \mid L \mid P \mid M \mid LP \mid LM \mid PM \mid LPM$$

$$L \rightarrow aLb \mid ab$$

$$M \rightarrow aMc \mid ac$$

$$P \rightarrow bPc \mid bc$$

Paso 3: Eliminar proyecciones unitarias

$$E \rightarrow ad \mid aEd \mid bJd \mid bd \mid cKd \mid cd$$

$$J \rightarrow bJd \mid bd \mid cKd \mid cd$$

$$K \rightarrow cKd \mid cd$$

$$F \rightarrow ad \mid aFd \mid aLb \mid ab \mid aMc \mid ac \mid bPc \mid bc \mid LP \mid LM \mid PM \mid LPM$$

$$L \rightarrow aLb \mid ab$$

$$M \rightarrow aMc \mid ac$$

$$P \rightarrow bPc \mid bc$$

$$S \rightarrow ad \mid bcl \mid ac \mid ab \mid aEd \mid bJd \mid cKd \mid aFd \mid aLb \mid aMc$$

$$bPc \mid LP \mid LM \mid PM \mid LPM$$